

Project VMAM 2025–2026

Theoretical answers

Casals Diego, Lainé Martin

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1 Direct problem

1.1 Variational formulation

The boundary conditions are:

- Left and right boundaries (Γ_{side}): homogeneous Neumann condition $\partial_n T = 0$ (insulated walls).
- Top boundary (Γ_{top}): Dirichlet condition $T = T_c = 343 \text{ K}$.
- Bottom boundary (Γ_{bottom}): Robin condition $\partial_n T + h(T - T_e) = 0$ where $T_e = 293 \text{ K}$ and $h = 1$ (door).

The temperature T inside the oven is given by the stationary heat equation:

$$-\text{div}(k \nabla T) = f \quad (1)$$

where k equals 1 inside S and 10 otherwise, and f is the heat source term that has the form $f = \sum_{i=1}^6 \alpha_i \mathbb{1}_{C_i}$.

We introduce a lifting function $\tilde{T} \in H^1(\Omega)$ such that $\tilde{T}|_{\Gamma_{\text{top}}} = T_c$. Then, we write $T = \tilde{T} + T_0$, where T_0 satisfies the homogeneous Dirichlet condition on Γ_{top} .

By multiplying Eq. (1) by a test function $v \in V = \{v \in H^1(\Omega) \mid v|_{\Gamma_{\text{top}}} = 0\}$ and integrating by parts, we obtain:

$$\int_{\Omega} k \nabla T \cdot \nabla v \, dx - \int_{\partial\Omega} k \partial_n T v \, d\sigma = \int_{\Omega} f v \, dx. \quad (2)$$

By substituting the boundary conditions (zero Neumann on Γ_{side} , Robin on Γ_{bottom} , and $v = 0$ on Γ_{top}) and adding the contribution of the lifting function, we arrive at the variational formulation:

$$\boxed{\text{Find } T_0 \in V \text{ such that } a(T_0, v) = l(v), \quad \forall v \in V} \quad (3)$$

where:

$$a(T_0, v) = \int_{\Omega} k \nabla T_0 \cdot \nabla v \, dx + \int_{\Gamma_{\text{bottom}}} k h T_0 v \, d\sigma, \quad (4)$$

$$l(v) = \int_{\Omega} f v \, dx + \int_{\Gamma_{\text{bottom}}} k h (T_e - \tilde{T}) v \, d\sigma - \int_{\Omega} k \nabla \tilde{T} \cdot \nabla v \, dx \quad (5)$$

Analysis (Lax-Milgram Theorem) Let $V = \{v \in H^1(\Omega) \mid v = 0 \text{ on } \Gamma_{\text{top}}\}$ be equipped with the H^1 -norm $\|v\|_{H^1}^2 = \|v\|_{L^2}^2 + \|\nabla v\|_{L^2}^2$. Recall that the thermal conductivity $k \in L^\infty(\Omega)$ satisfies $1 \leq k(\mathbf{x}) \leq 10$ a.e. in Ω , and $h = 1$.

To apply the Lax-Milgram theorem, we must verify three hypotheses:

1. **Continuity of the bilinear form $a(\cdot, \cdot)$:** For all $u, v \in V$, using the triangle inequality and the upper bound of k , we have:

$$|a(u, v)| \leq 10 \left| \int_{\Omega} \nabla u \cdot \nabla v \, dx \right| + 10 \left| \int_{\Gamma_{\text{bottom}}} u v \, d\sigma \right|.$$

By applying Cauchy-Schwarz, and using the Trace inequality for the right integral, we obtain:

$$|a(u, v)| \leq 10 \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + 10 C_{\text{tr}}^2 \|u\|_{H^1} \|v\|_{H^1}.$$

Since $\|\nabla v\|_{L^2} \leq \|v\|_{H^1}$, there exists a constant $M > 0$ such that $|a(u, v)| \leq M \|u\|_{H^1} \|v\|_{H^1}$. Thus, $a(\cdot, \cdot)$ is continuous.

2. **Coercivity of $a(\cdot, \cdot)$:** For all $v \in V$, since $k \geq 1$ and $khv^2 \geq 0$, we have:

$$a(v, v) = \int_{\Omega} k |\nabla v|^2 dx + \int_{\Gamma_{\text{bottom}}} khv^2 d\sigma \geq \int_{\Omega} |\nabla v|^2 dx = \|\nabla v\|_{L^2}^2.$$

Since Ω is connected and the Dirichlet boundary Γ_{top} has a strictly positive measure, the Poincaré inequality holds on V .

Thus, there exists a constant $C_P > 0$ such that $\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)}$ for all $v \in V$.

Using the definition of the H^1 -norm ($\|v\|_{H^1}^2 = \|v\|_{L^2}^2 + \|\nabla v\|_{L^2}^2$), this implies that there exists $\alpha > 0$ such that:

$$a(v, v) \geq \|\nabla v\|_{L^2}^2 \geq \alpha \|v\|_{H^1}^2.$$

Thus, the bilinear form $a(\cdot, \cdot)$ is coercive on V .

3. **Continuity of the linear form $l(\cdot)$:** For all $v \in V$, we use the triangle inequality and the bounds of k :

$$|l(v)| \leq \left| \int_{\Omega} f v dx \right| + 10 \left| \int_{\Gamma_{\text{bottom}}} (T_e - \tilde{T}) v d\sigma \right| + 10 \left| \int_{\Omega} \nabla \tilde{T} \cdot \nabla v dx \right|.$$

Applying the Cauchy-Schwarz inequality and the Trace inequality:

$$|l(v)| \leq \|f\|_{L^2} \|v\|_{L^2} + 10 \|T_e - \tilde{T}\|_{L^2(\Gamma_{\text{bottom}})} C_{\text{tr}} \|v\|_{H^1} + 10 \|\nabla \tilde{T}\|_{L^2} \|\nabla v\|_{L^2}.$$

Since $\tilde{T}$ is a fixed function in $H^1(\Omega)$ and $f \in L^2(\Omega)$, all terms multiplying the norms of v are bounded constants. Hence, there exists $C > 0$ such that:

$$|l(v)| \leq C \|v\|_{H^1}.$$

Thus, $l(\cdot)$ is continuous.

Conclusion: Since V is a Hilbert space, $a(\cdot, \cdot)$ is a continuous and coercive bilinear form, and $l(\cdot)$ is a continuous linear form, the Lax-Milgram theorem guarantees the existence and uniqueness of the solution $T_0 \in V$.

1.2 Finite element approximation

We approximate V by a finite-dimensional space $V_h \subset V$, based on a regular triangular mesh $\mathcal{T}_h$ of Ω using $\mathbb{P}_1$ finite elements. Let $(\varphi_j)_{j=1}^N$ be the nodal basis functions associated with the nodes that are not on the Dirichlet boundary (i.e., strictly vanishing on Γ_{top}).

We approximate the unknown T_0 by $T_{0,h} = \sum_{j=1}^N U_j \varphi_j \in V_h$, where the U_j are the unknown temperatures at the nodes. Similarly, the continuous lifting function is discretized as $\tilde{T}_h$, which takes the fixed value T_c at the nodes located on Γ_{top} and vanishes at all other nodes.

By substituting $T_{0,h}$ into the continuous variational formulation of Eq. (3) and choosing the test functions $v = \varphi_i$ for $i = 1, \dots, N$, we obtain the linear system:

$$\mathbf{A} \mathbf{U} = \mathbf{b}, \tag{6}$$

where the elements of the matrix $\mathbf{A}$ and the right-hand side vector $\mathbf{b}$ are directly deduced from the bilinear and linear forms:

$$A_{ij} = a(\varphi_j, \varphi_i) = \int_{\Omega} k \nabla \varphi_j \cdot \nabla \varphi_i dx + \int_{\Gamma_{\text{bottom}}} kh \varphi_j \varphi_i d\sigma, \tag{7}$$

$$b_i = l(\varphi_i) = \int_{\Omega} f \varphi_i dx + \int_{\Gamma_{\text{bottom}}} kh (T_e - \tilde{T}_h) \varphi_i d\sigma - \int_{\Omega} k \nabla \tilde{T}_h \cdot \nabla \varphi_i dx. \tag{8}$$

The matrix $\mathbf{A}$ is symmetric, positive-definite, and sparse. Once the system is solved for the vector $\mathbf{U}$, the full approximate temperature distribution in the oven is reconstructed by adding back the discrete lifting function:

$$T_h(\mathbf{x}) = \sum_{j=1}^N U_j \varphi_j(\mathbf{x}) + \tilde{T}_h(\mathbf{x}). \quad (9)$$

Remark: In our numerical implementation with *FreeFem++*, the construction of $\tilde{T}_h$ and the transfer of the lifting terms to the right-hand side $\mathbf{b}$ are handled automatically, so we directly solve for the full temperature T_h without explicitly separating the lifting function. We implement the equation on *FreeFem++* as:

$$\int_{\Omega} k \nabla T \cdot \nabla v \, dx + \int_{\Gamma_{bottom}} kh T v \, d\sigma = \int_{\Omega} f v \, dx + \int_{\Gamma_{bottom}} kh T_e v \, d\sigma \quad (10)$$

1.3 FreeFem++ implementation and comments

The numerical simulation (see `directPb.edp`) of the direct problem yields the temperature distribution inside the oven for a uniform set of source intensities ($\alpha_i = 10^4$), see Fig. 1.

Temperature distribution on the oven with fixed resistors

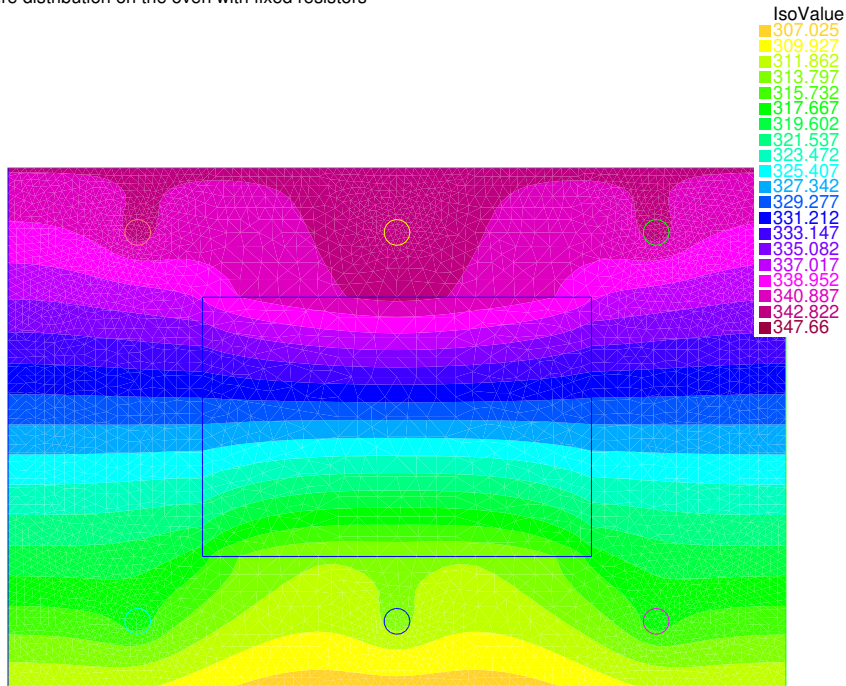


Figure 1: Temperature distribution inside the oven for the direct problem ($\alpha_i = 10^4$).

From the obtained temperature map (Fig. 1), several key physical behaviors can be observed. Firstly, the top wall boundary condition is correctly enforced, as the temperature at the top boundary is uniformly equal to T_c . The lower boundary displays significant heat loss towards the external environment at temperature T_e , which is perfectly consistent with the Robin boundary condition. On the left and right insulated boundaries (homogeneous Neumann conditions), the

isotherms are strictly perpendicular to the walls. This graphically confirms that the normal heat flux is zero.

Then, the region S is not heated uniformly, since the top of the piece to cook is warmer than the bottom. So, we don't have a good temperature distribution for cooking, which motivates the optimization problem in Sec. 2, in order to have different source intensities α_i that can yield a more uniform temperature distribution in S .

1.4 Description of the oven warm up phase

To model the heating phase of the oven from an initial cold state, we introduce the time variable $t \in [0, T]$. The temperature distribution $T(\mathbf{x}, t)$ is governed by the transient heat equation, which includes the thermal inertia term ρc (volumetric heat capacity):

$$\rho c \frac{\partial T}{\partial t} - \operatorname{div}(k \nabla T) = f \quad \text{in } \Omega \times [0, T], \quad (11)$$

subject to the same boundary conditions as the stationary problem. Assuming normalized physical properties for the air inside the oven, we set $\rho c = 1$. Initially, the oven is at room temperature:

$$T(\mathbf{x}, 0) = T_e \quad \text{in } \Omega. \quad (12)$$

We introduce a time step $\delta t > 0$ and denote $t_n = n\delta t$. To ensure unconditional numerical stability, we use the first-order implicit Euler scheme, approximating the time derivative by $\partial_t T^{n+1} \approx \frac{T^{n+1} - T^n}{\delta t}$.

Following the stationary case procedure, we introduce the bilinear form $m(u, v) = \int_{\Omega} u v \, dx$. By isolating the unknown terms at time t_{n+1} on the left-hand side, we obtain the fully discrete variational formulation at each time step:

For $n \geq 0$, find $T_h^{n+1} \in V_h$ such that $\forall v_h \in V_h$,

$$m(T_h^{n+1}, v_h) + \delta t a(T_h^{n+1}, v_h) = m(T_h^n, v_h) + \delta t l(v_h),$$

(13)

where $a(\cdot, \cdot)$ and $l(\cdot)$ are the forms defined in Eq. (3). As $t \rightarrow +\infty$, this transient solution naturally converges toward the stationary solution.

At the final time step (here $T = 0.5$), we obtain (see `transientPb.edp`) a temperature distribution that is close to the stationary solution, which validates the implementation of the transient scheme (see Fig. 2).

Temperature distribution on the oven with fixed resistors at t=0.5

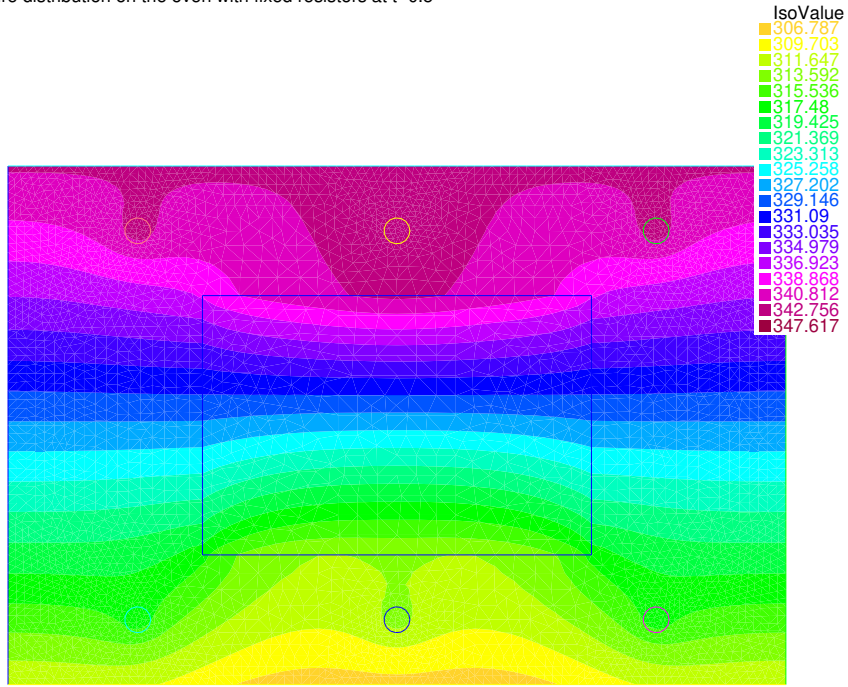


Figure 2: Temperature distribution inside the oven for the transient direct problem ($\alpha_i = 10^4$) at the final time step.

The warm-up phase can be visualized as a sequence of temperature maps at different time steps, showing how the oven heats up from the initial cold state to the steady-state distribution.

2 Optimization problem

2.1 Linear decomposition of the solution

Let $T(\boldsymbol{\alpha}, \mathbf{x})$ be the solution of the problem.

First, we define $T_0(\mathbf{x})$ as the solution to the physical problem without any internal heat sources ($f = 0$), but subjected to the exact same external boundary conditions as T :

$$\begin{cases} -\operatorname{div}(k\nabla T_0) = 0 & \text{in } \Omega, \\ T_0 = T_c & \text{on } \Gamma_{\text{top}}, \\ \partial_n T_0 + hT_0 = hT_e & \text{on } \Gamma_{\text{bottom}}, \\ \partial_n T_0 = 0 & \text{on } \Gamma_{\text{side}}. \end{cases} \quad (14)$$

Next, we introduce the residual temperature field $U(\boldsymbol{\alpha}, \mathbf{x}) = T(\boldsymbol{\alpha}, \mathbf{x}) - T_0(\mathbf{x})$. By subtracting the system of equations for T_0 from the system for T , and by linearity of the differential operators, U satisfies:

$$\begin{cases} -\operatorname{div}(k\nabla U) = \sum_{i=1}^6 \alpha_i \mathbb{1}_{C_i} & \text{in } \Omega, \\ U = 0 & \text{on } \Gamma_{\text{top}}, \\ \partial_n U + hU = 0 & \text{on } \Gamma_{\text{bottom}}, \\ \partial_n U = 0 & \text{on } \Gamma_{\text{side}}. \end{cases} \quad (15)$$

Since the partial differential operator $-\operatorname{div}(k\nabla \cdot)$ along with these homogeneous boundary conditions forms a purely linear operator, the solution U depends linearly on the source term $\sum_{i=1}^6 \alpha_i \mathbb{1}_{C_i}$.

By the superposition principle, we can therefore define $T_i(\mathbf{x})$ for each $i \in \{1, \dots, 6\}$ as the elementary solution of Eq. (15):

$$\begin{cases} -\operatorname{div}(k\nabla T_i) = \mathbb{1}_{C_i} & \text{in } \Omega, \\ T_i = 0 & \text{on } \Gamma_{\text{top}}, \\ \partial_n T_i + hT_i = 0 & \text{on } \Gamma_{\text{bottom}}, \\ \partial_n T_i = 0 & \text{on } \Gamma_{\text{side}}. \end{cases} \quad (16)$$

Due to linearity, it immediately follows that $U(\boldsymbol{\alpha}, \mathbf{x}) = \sum_{i=1}^6 \alpha_i T_i(\mathbf{x})$. Substituting this back into $T = T_0 + U$, we obtain the decomposition of the temperature field:

$$T(\boldsymbol{\alpha}, \mathbf{x}) = T_0(\mathbf{x}) + \sum_{i=1}^6 \alpha_i T_i(\mathbf{x}). \quad (17)$$

where T_0 follows Eq. (14) and each T_i follows Eq. (16).

2.2 Reduction to a linear system

We seek the optimal vector $\boldsymbol{\alpha}^* = (\alpha_1^*, \dots, \alpha_6^*)^T$ that minimizes the cost function:

$$J(\boldsymbol{\alpha}) = \frac{1}{2} \int_S |T(\boldsymbol{\alpha}, \mathbf{x}) - T_s|^2 d\mathbf{x}. \quad (18)$$

By substituting the linear decomposition from Eq. (17), J becomes a convex quadratic function of $\boldsymbol{\alpha}$:

$$J(\boldsymbol{\alpha}) = \frac{1}{2} \int_S \left(T_0(\mathbf{x}) + \sum_{i=1}^6 \alpha_i T_i(\mathbf{x}) - T_s \right)^2 d\mathbf{x}. \quad (19)$$

The necessary and sufficient first-order optimality conditions for this unconstrained convex minimization problem are $\frac{\partial J}{\partial \alpha_j}(\boldsymbol{\alpha}^*) = 0$ for all $j \in \{1, \dots, 6\}$. Differentiating J gives:

$$\int_S \left(T_0(\mathbf{x}) + \sum_{i=1}^6 \alpha_i^* T_i(\mathbf{x}) - T_s \right) T_j(\mathbf{x}) \, d\mathbf{x} = 0, \quad \forall j \in \{1, \dots, 6\}. \quad (20)$$

By separating the terms that depend on $\boldsymbol{\alpha}^*$ from the known quantities, we obtain the following 6×6 linear system:

$$M \boldsymbol{\alpha}^* = \mathbf{b}, \quad \text{with } M_{ij} = \int_S T_i T_j \, d\mathbf{x}, \quad \text{and } b_j = \int_S (T_s - T_0) T_j \, d\mathbf{x}. \quad (21)$$

M is symmetric by construction. Furthermore, for a vector $\mathbf{v} = (v_1, \dots, v_6)^T$, we have:

$$\mathbf{v}^T M \mathbf{v} = \sum_{i=1}^6 \sum_{j=1}^6 v_i \left(\int_S T_i(\mathbf{x}) T_j(\mathbf{x}) \, d\mathbf{x} \right) v_j = \int_S \left(\sum_{i=1}^6 v_i T_i(\mathbf{x}) \right)^2 \, d\mathbf{x} \geq 0, \quad (22)$$

which proves that M is positive semi-definite. Assuming the localized thermal footprints $(T_i|_S)_{i=1}^6$ are linearly independent over S (which is the case here), the integral vanishes if and only if $\mathbf{v} = \mathbf{0}$. Therefore, M is positive definite, which guarantees the invertibility of the linear system Eq. (21) and the uniqueness of the optimal control $\boldsymbol{\alpha}^*$.

2.3 Optimal coefficients and comments

The resolution of the linear system Eq. (21) with $T_s = 400 \text{ K}$ yields the optimal control vector $\boldsymbol{\alpha}^*$, which dictates the power intensity of each individual resistor. The computed optimal values are (see `optimalPb.edp`):

$$\begin{array}{lll} \alpha_1^* = 159\,601, & \alpha_2^* = 168\,331, & \alpha_3^* = 158\,176, \\ \alpha_4^* = 502\,764, & \alpha_5^* = 412\,308, & \alpha_6^* = 497\,148. \end{array}$$

The resulting optimized temperature distribution is displayed in Fig. 3.

Temperature distribution on the oven with optimal resistors values

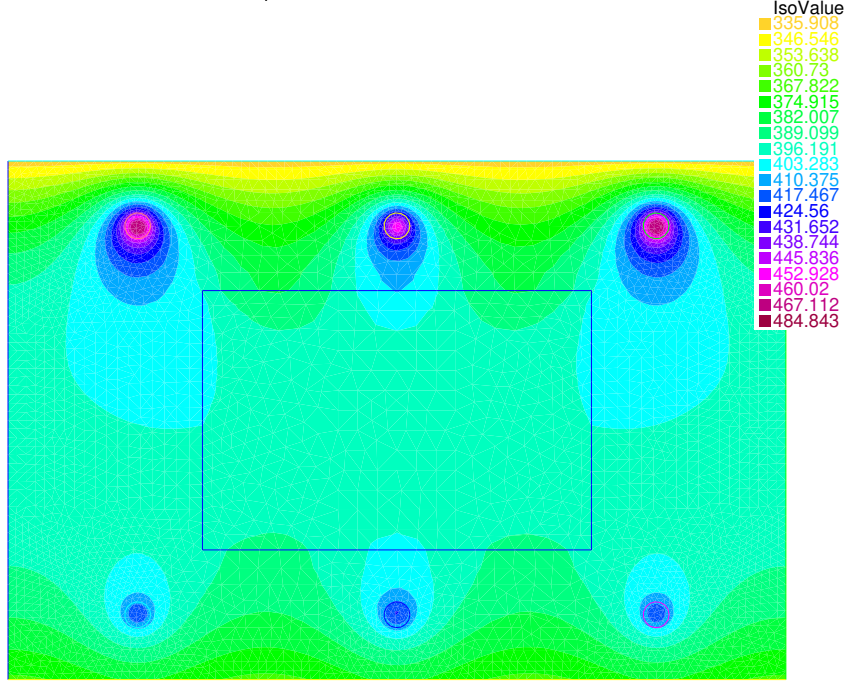


Figure 3: Temperature distribution inside the oven computed with the optimal heat sources α^* .

The objective of the optimization is successfully met, as the residual L^2 -error over the cooking region S drops to a low value of 3.06. Visually, the temperature field within S (the central rectangle) is almost uniform.

We can also remark that the optimization algorithm naturally accounted for the asymmetric boundary conditions of the oven. The top resistors ($\alpha_4^*, \alpha_5^*, \alpha_6^*$) must operate at a significantly higher intensity ($\approx 5 \times 10^5$) compared to the bottom resistors ($\alpha_1^*, \alpha_2^*, \alpha_3^* \approx 1.6 \times 10^5$). This perfectly balances the distinct heat dissipation rates of the upper and lower walls (Dirichlet condition is more dissipative than Robin condition).

Furthermore, the resistors located closer to the center of S (α_2^* and α_5^*) have different values than the ones located on the edges (one is lower and one is higher).

2.4 Design of an improved oven configuration

While the previous optimization yielded a satisfactory L^2 -error, we can have better results. By modifying the location, shape, and number of the heat sources, we propose a new configuration aimed at achieving a more naturally uniform heat distribution.

So, we chose to increase the number of resistors from 6 to 8. We distribute them symmetrically around the cooking zone: two on the top, two on the bottom, two on the left, and two on the right.

Moreover, the circular point sources are replaced by rectangular resistors, in order to follow the shape of the cooking region.

We implemented this new configuration in FreeFem++ (see `optimalPbBetterOven.edp`) and solved the optimization problem using the same approach as Sec. 2.2.

As shown in Fig. 4, the new optimal temperature distribution is remarkably homogeneous.

Temperature distribution with 8 rectangular resistors

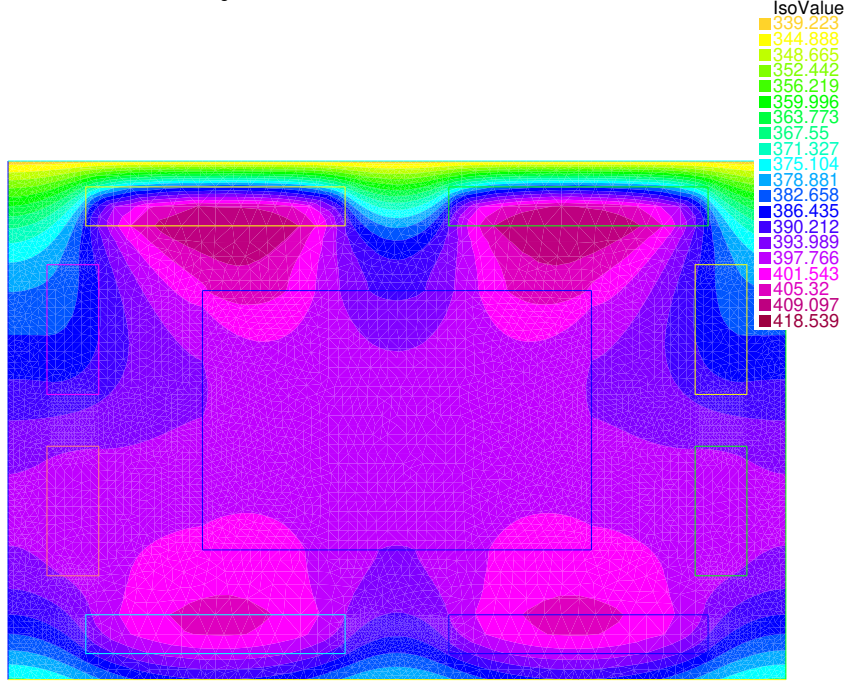


Figure 4: Temperature distribution for the proposed 8-strip configuration.

The residual L^2 -error drops significantly from 3.06 to 2.167. Furthermore, the optimal power coefficients α^* require lower peak intensities, proving that this design is also far more energy-efficient and realistic for industrial manufacturing.

3 Inverse problem

In this section, we study an anomaly corresponding to a single unknown high-intensity heat source $f = \alpha \mathbf{1}_C$. Both its intensity α and its physical location (x_c, y_c) (the center of the circular region C of radius r_0) are unknown parameters that must be identified.

To perform this detection, we rely on temperature measurements T^{mes} taken on the lateral boundaries of the oven Γ_{side} .

For a fixed position (x_c, y_c) , we seek the intensity α that minimizes the L^2 -error between our model prediction and the measurements on Γ_{side} :

$$\mathcal{J}(x_c, y_c, \alpha) = \frac{1}{2} \int_{\Gamma_{\text{side}}} |T(x_c, y_c, \alpha) - T^{\text{mes}}|^2 d\sigma. \quad (23)$$

Minimizing this functional directly with respect to both the spatial coordinates and the intensity is a difficult problem. To overcome this difficulty, we apply the exact same strategy as in Sec. 2. Using the linearity of the stationary heat equation, the temperature field can be decomposed using the superposition principle:

$$T(x_c, y_c, \alpha) = T_0 + \alpha \tilde{T}(x_c, y_c), \quad (24)$$

where T_0 is the solution to Eq. (14) (without any anomaly) and $\tilde{T}$ follows Eq. (16) (with a unit source ($\alpha = 1$) strictly located at (x_c, y_c)).

The necessary optimality condition $\frac{\partial \mathcal{J}}{\partial \alpha} = 0$ provides an explicit analytical formula for the optimal intensity associated with any given position:

$$\alpha^*(x_c, y_c) = \frac{\int_{\Gamma_{\text{side}}} (T^{\text{mes}} - T_0) \tilde{T} d\sigma}{\int_{\Gamma_{\text{side}}} \tilde{T}^2 d\sigma}. \quad (25)$$

By injecting $\alpha^*(x_c, y_c)$ back into Eq. (23), the cost functional depends solely on the spatial coordinates.

We proceed with a grid search over a 30×23 spatial grid covering the domain $[-1.45, 1.45] \times [-0.95, 0.95]$. For each grid point (x_k, y_k) :

1. We solve the direct problem to obtain $\tilde{T}(x_k, y_k)$ using Eq. (16).
2. We compute the optimal intensity $\alpha^*(x_k, y_k)$ using Eq. (25).
3. We evaluate the reduced cost functional $\mathcal{J}(x_k, y_k, \alpha^*)$.

The grid point that minimizes $\mathcal{J}$ is identified as the anomaly's location, and its associated α^* is the detected intensity.

We implemented this approach in `inversePb.edp`. For different values of the true anomaly parameters (x_c, y_c, α) , we successfully detected the anomaly with good accuracy. A slight discrepancy remains, which is inherently tied to the spatial discretization limit of the grid. While a nested two-pass strategy (performing a localized second grid search around the initially detected position) could further reduce this error, the single-pass high-resolution grid already yields highly satisfactory accuracy while maintaining a reasonable computational time.